



13

Explanation

13.1 Knowing Why

What does science do for us? In chapter 12 I argued for a version of scientific realism, according to which one aim of science is describing the real structure of the world. Science aims to tell us, and often succeeds in telling us, what the world is like. But it is also common to think that science tells us *why* things happen; we learn from science not just what goes on but why it does. Science apparently seeks to *explain* as well as describe. So we seem to face a new question. What *is* it for a scientific theory to explain something? In what sense does science give us an *understanding* of phenomena, as opposed to mere descriptions of what there is and what happens?

The idea that science aims at explanations of *why* things happen has sometimes aroused suspicion in philosophers, and it has also done so in scientists themselves. Such distrust is reasonably common within strong empiricist views. Empiricists have often seen science, most fundamentally, as a system of rules for predicting experience. When explanation is put forward as an *extra* goal for scientific theories, empiricists get nervous.

There is a complicated relationship between this problem of explanation and the problem of analyzing confirmation and evidence (chapters 3, 14). The hope has often been to treat these problems separately. Understanding evidence is problem 1; this is the problem of analyzing what it is to have evidence to believe that a scientific theory is true. Understanding explanation is problem 2; here we assume that we have already chosen our scientific theories, at least for now. We want to work out how our theories provide explanations. In principle, we can make a distinction of this kind. But there is a close connection between the issues. The solution to problem 2 may affect how we solve problem 1. Theories are often preferred by scientists because they seem to yield good explanations of puzzling phenomena. In chapter 3, *explanatory inference* was defined as inference from a set of data to a hypothesis about a structure or process that would explain the

data. This seems much more common in science than the traditional philosophical idea of inductive inference (inference from particular cases to generalizations). This suggests that there is a close relation between the problem of analyzing explanation and the problem of analyzing evidence.

There is a very large literature on explanation, but these issues will get a whirlwind treatment in this book. One reason for this is that I think the philosophy of science has approached the problem of explanation in a mistaken way. To some extent, that is true of many topics in this book; there have been plenty of wrong turns in the philosophy of science. But in the case of explanation, I think the error has been fairly clear; I will describe it in section 13.3. So some of the views presented in this chapter are rather unorthodox.

13.2 The Rise and Fall of the Covering Law Theory of Explanation

Empiricist philosophers, I said above, have sometimes been distrustful of the idea that science explains things. Logical positivism is an example. The idea of explanation was sometimes associated by the positivists with the idea of achieving deep metaphysical insight into the world—an idea they would have nothing to do with. But the logical positivists and logical empiricists did make peace with the idea that science explains. They did this by construing “explanation” in a low-key way that fitted into their empiricist picture.

The result was the *covering law* theory of explanation. This was the dominant philosophical theory about scientific explanation for a good part of the twentieth century. The view is now dead, but its rise and fall are instructive.

The covering law theory of explanation was first developed in detail by Carl Hempel and Paul Oppenheim in a paper (1948) that became a centerpiece of logical empiricist philosophy. Let us begin with some terminology. In talking about how explanation works, the *explanandum* is whatever is being explained. The *explanans* is the thing that is doing the explaining. If we ask “why X?” then X is the explanandum. If we answer “because Y,” then Y is the explanans.

The basic ideas of the covering law theory are simple. Most fundamentally, to explain something is to *show how to derive it in a logical argument*. The explanandum will be the conclusion of the argument, and the premises are the explanans. A good explanation must first of all be a good logical argument, but in addition, the premises must contain at least one statement of a *law of nature*. The law must make a real contribution to the argument; it cannot be something merely tacked on. (Of course, for an explanation to

be a good one in the fullest sense, the premises must also be *true*. But the first task here is to describe what sort of statements *would* give us a good explanation of a phenomenon, if the statements were true.)

Some explanations (both in science and in everyday life) are of particular events, while others are directed at general phenomena or regularities. For example, we might try to explain the particular fact that the U.S. stock market crashed in 1929, in terms of economic laws operating against the background conditions of the day. And we can also explain patterns; Newton is often seen as giving an explanation of Kepler's laws of planetary motion in terms of more basic laws of mechanics in combination with assumptions about the layout of the solar system. In both cases, the covering law theory sees these explanations as expressible in terms of arguments from premises to conclusions. Some of the arguments that express explanations will be deductively valid, but this is not required in all cases. The covering law theory was intended to allow that some good explanations could be expressed as nondeductive arguments ("inductive" arguments, in the logical empiricists' broad sense of the term). If we can take a particular phenomenon and embed it within an argument in which the premises include a law and bestow high probability on the conclusion, this yields a good explanation of the phenomenon.

There were many problems of detail encountered in attempts to formulate the covering law theory precisely (Salmon 1989). The problems were more difficult in the case of nondeductive arguments, and also in the case of explaining patterns rather than particular events. I won't worry about the technicalities here. The basic idea of the covering law theory is simple and clear: to explain something is to show how to derive it in a logical argument of a kind that makes use of a law in the premises. To explain something is to *show that it is to be expected*, to show that it is *not surprising*, given our knowledge of the laws of nature.

For the covering law theory, there is not much difference between explanation and prediction. To predict something, we put together an argument and try to show that it is to be expected, though we don't know for sure yet whether it is going to happen. When we explain something, we know that it has happened already, and we show that it *could* have been predicted, using an argument containing a law. You might be wondering at this point what a "law of nature" is supposed to be. This was a troubling topic for logical empiricism and has continued to be troubling for everyone else. But a "law of nature" was not supposed to be something very grandiose. It was supposed to be a kind of basic regularity, a basic pattern, in the flow of events. (I return to this question in section 13.4.)

Though I use the phrase "covering law theory" here, another name for the

theory is the “D-N theory,” or “D-N model,” of explanation. “D-N” stands for “deductive-nomological,” where the word “nomological” is from the Greek word for law, *nomos*. The term “D-N” can be confusing because, as I said, the argument in a good explanation need not be deductive. So “D-N” really only refers to some covering law explanations, the deductive ones.

That concludes my sketch of the covering law theory. I now move on to what is wrong with it. This is a case where we have something close to a knockdown argument. Although there are many famous problems with the covering law theory, the most convincing problem is usually called the *asymmetry problem*. And the most famous illustration of the asymmetry problem is the case of the flagpole and the shadow.

Suppose we have a flagpole casting a shadow on a sunny day. Someone asks: why is the shadow X meters long? According to the covering law theory, we can give a good explanation of the shadow by deducing the length of the shadow from the height of the flagpole, the position of the sun, the laws of optics, and some basic trigonometry. We can show why that length of shadow was to be expected, given the laws and the circumstances. The argument can even be made deductively valid. So far, so good. The problem is that we can run just as good an argument in another direction. Just as we can deduce the length of the shadow from the height of the pole (plus optics and trigonometry), *we can deduce the height of the pole from the length of the shadow* (and the same laws). An equally good *argument*, logically speaking, can be run in both directions; either can give information about the other. But it seems that we cannot run an equally good *explanation* in both directions, though the covering law theory says we can. It is fine to explain the length of the shadow in terms of the flagpole and the sun, but it is not fine to explain the length of the flagpole in terms of the shadow and the sun. (At least, it is not fine unless this is a very unusual flagpole—perhaps one that is designed to regulate its own length to maintain a particular shadow.)

What we find here is that explanations have a kind of directionality. Some arguments (though not all) can be reversed and remain good as arguments. But explanations cannot be reversed in this way (except in some special cases). So not all good arguments that contain laws are good explanations. This objection to the covering law theory was famously given (using a slightly different example) by Sylvain Bromberger (1966).

Once this point is seen, it becomes obvious and devastating. The covering law theory sees explanation as very similar to prediction; the only difference is what you know and what you don’t know. But this is a mistake. Consider the concept of a *symptom*. Symptoms can be used to predict, but they cannot be used to explain. Yet a symptom can often be used in a good logical argument, along with a law, to show that something is to be expected.

If you know that only disease D produces symptom S, then you can make inferences from S to D. You might in some cases be able to make predictions from D to S, too. But you cannot *explain* a disease in terms of a symptom. Explanation only runs one way, from D to S, no matter how many different kinds of inferences can be made in other directions. And, further, it seems that good explanations of S can be given in terms of D even if S is *not* a very reliable symptom of D, even if S is not always to be expected when someone has D. This is a separate problem for the covering law theory, often discussed using the example of some unreliable but unpleasant symptoms of syphilis.

In some of these cases, the covering law theory can engage in fancy footwork to evade the problem. But other cases, including the original flagpole case, seem immune to footwork. Hempel's own attitude to the issue was puzzling. He actually anticipated the problem, but dismissed it (Hempel 1965, 352–54). His strategy was to accept that if his theory allowed explanations to run in two directions in cases where it seems that explanation only runs in one direction, then both directions must really be OK. In some actual scientific cases this reply seems reasonable; there are cases in physics where it *is* hard to tell which direction(s) the explanation(s) are running in. But in other cases the direction seems completely clear. In the case of the flagpole and the shadow, this reply seems hopeless.

There are other good arguments against the covering law theory (Salmon 1989), but the asymmetry problem is the killer. It also seems to be pointing us toward a better account of explanation.

13.3 Causation, Unification, and More

What is it about the flagpole's height that makes it a good explanation for the length of the shadow, and not vice versa? The answer seems straightforward. The shadow is *caused* by the interaction between sunlight and the flagpole. That is the direction of causation in this case, and that is the direction of explanation also. So we seem to get an immediate suggestion from the flagpole case for how to build a better theory: to explain something is to describe what caused it. Why did the dinosaurs become extinct 65 million years ago? Here again, our request for an explanation seems to amount to a request for information about what caused the extinction.

Although that conclusion seems compelling, it has not been universally accepted, and it raises many further problems. The biggest question raised at this point is, What *is* causation? We confidently used the idea of causation to resolve the flagpole case, but the whole idea of causation and causal connection is extremely controversial in philosophy. For many philoso-

phers, causation is a suspicious metaphysical concept that we do best to avoid when trying to understand science. This suspicion is, again, common within the empiricist tradition. It derives from the work of Hume. The suspicion is directed especially at the idea of causation as a sort of hidden connection between things, unobservable but essential to the operation of the universe. Empiricists have often tried to understand science without supposing that science concerns itself with alleged hidden connections of that kind. The rise of scientific realism in the latter part of the twentieth century led to some easing of this anxiety. But many philosophers would be pleased to see an adequate account of science that did not get entangled with issues about causation.

Despite this unease, toward the end of the twentieth century, the main proposal about explanation being discussed, in different ways, was the idea that explaining something is giving information about how it was caused. Some sophisticated analyses were developed that sought to use probability theory to clarify this basic idea (Salmon 1984; Suppes 1984). It might seem initially that this view of explanation is most directly applied to explanations of particular events (like the extinction of the dinosaurs), but it can also be applied to the explanation of patterns. We can ask, Why does inbreeding produce an increase in birth defects? The explanation will describe a general kind of causal process that is involved in producing the phenomenon (a process involving an increased chance that two copies of a recessive gene will be brought together in a single individual).

Claiming that causation is the key to explanation does not settle all the issues about explanation. We need to know what *kind* of information about causes is needed for a good explanation. One way to think of the situation is to imagine an idealized “complete” explanation that contains *everything* in the causal history of the event to be explained, specified in total detail (Railton 1981). No one ever wants to be told the complete explanation for a phenomenon, and we never know these complete explanations. Instead, in any context of discussion in which a request for an explanation is made, some *piece* or *pieces* of the complete explanation will be relevant. We are often able to know, and describe, these relevant pieces of a total causal structure. To give a good explanation in actual practice, all that is required is a description of these relevant pieces of the whole.

One main alternative to the analysis of explanation in terms of causation was developed in the years after the demise of the covering law theory. This was the idea that explanation should be analyzed in terms of *unification*. This idea was developed in detail by Michael Friedman (1974) and Philip Kitcher (1981, 1989). But as Kitcher also emphasized, the idea was actually present all along in logical empiricism. The idea that explanation

is unification was a sort of “unofficial” theory of explanation within much logical empiricism, in contrast to the “official” covering law theory (see, for example, Feigl 1943). This unofficial theory is a good deal better than the official theory. Often the two approaches were mixed in together; to show the connection between particular events and a general law is, after all, to achieve a kind of unification. Why not develop a theory of unification in science that is not tied to the idea of deriving phenomena from laws?

So the *unificationist* theory holds that explanation in science is a matter of connecting a diverse set of facts by subsuming them under a set of basic patterns and principles. Science constantly strives to reduce the number of things that we must accept as fundamental. We try to develop general *explanatory schemata* (explanatory schemes) that can be applied as widely as possible. This proposal certainly makes a lot of sense of how scientists operate. Indeed, it seems clear that what produces an “Aha!” reaction is often the realization that some odd-looking phenomenon is really a case of something more general. Kitcher also argues for this view with cases from the history of science. He argues that some very famous theories—Darwin’s theory of evolution and Newton’s later work on the nature of matter—were compelling to scientists in their early stages despite not making many specific new predictions, because they promised to *explain so much*. And this “explanatory promise” seems to have been the ability of those theories to *unify* a great range of phenomena with a few general principles.

In Kitcher’s case, another reason for developing the unificationist theory was a distrust of the idea of causation. This led Kitcher to try, for some years, to develop a theory of explanation *entirely* in terms of unification. But what about the flagpole and the shadow, and the asymmetry in which can explain which? Kitcher argued that we do tend to describe this asymmetry in causal terms, but this causal talk is really a loose summary of more basic asymmetries that involve unification (Kitcher 1989).

So we have two main proposals to replace the covering law theory: the causal theory and the unification theory. These have often been treated as competitors: “Does causation win or does unification win?” But this is surely a mistake. We do not have to choose. Again, beware the dubious allure of simplicity in philosophical theories! Much of the time, to explain something is to describe the causal mechanisms behind it or the causal history leading up to it. That is true *much* of the time, but there is no need to hold that it is true *all* the time. In some cases there can be pretty clear explanatory relations between patterns or principles, even when causal language is hard to apply to the situation. Often this seems to involve unification. Nothing stops us from holding that a variety of different relations can be explanatory.

Recently, ideas similar to this have been emerging in the philosophy of science. Wesley Salmon was for many years one of the main partisans for the idea that causation is the key to explanation. But he eventually accepted that unification is also part of the story. Sometimes he seemed to think of causation and unification as two sides of the same explanatory coin, and some other times as alternative explanatory projects (1989, 1998). Kitcher, who tried for years to avoid using the idea of causation to analyze explanation, instead telling the whole story in terms of unification, has now decided that this was probably a mistake and the concept of causation is not so dubious after all (personal communication, 2002).

So what might be emerging is a kind of “pluralism” about explanation in the philosophy of science. This is a step in the right direction, but I suggest that the whole issue has been approached wrongly. (This is where I become unorthodox.) The most peculiar thing about the discussion of explanation by philosophers has been the assumption that explanation is the kind of thing that requires analysis in terms of a single special relation or a short list of special relations. It is a mistake to think there is one basic relation that is *the* explanatory relation (as in the covering law theory, the causal theory, and the unification theory), and it is also a mistake to think that there are some definite two or three such relations.

The alternative view is to recognize that the idea of explanation operates differently within different parts of science—and differently within the same part of science at different times. The word “explanation” is used in science for something that is sought, and sometimes achieved, by the development of theories, but what exactly is being sought is not constant in all of science. And we cannot get the right analysis by claiming that within all of science, a good explanation is something that satisfies *either* the causal test *or* the unification test (etc.). This familiar form of “pluralism” leaves out the way that different scientific fields will establish definite criteria for what will pass as a good explanation. The standards for a good explanation in field A need not suffice in field B. If an “ism” is required, the right analysis of explanation is a kind of *contextualism*—a view that treats the standards for good explanation as partially dependent on the scientific context.

Kuhn argued some years ago for a view of this kind (1977a). In a paper about the history of physics, he claimed that different theories (or paradigms) tend to bring with them their own standards for what counts as a good explanation. He argued, further, that standards about whether a relation counts as “causal” also depend on paradigms. The concepts of explanation and causation are, to some extent at least, internal to different scientific fields and historical periods.

In the case of causation, a philosopher might reply to Kuhn, with some

justification, that just because different people have *thought* differently about what causation is does not mean that there *is* no fact of the matter. Fair enough (at least for now). But in the case of explanation, I think this reply has little force. If two scientific fields single out different relations and call them “explanation,” there need be no factual error that one or the other is making.

To support this claim, Kuhn focused (as he did in *Structure*) on the case of Newton’s theory of gravity. Does Newton’s theory *explain* the falling of objects, given that Newton’s treatment of gravity gave no intuitive mechanism but only a mathematical relationship? Some answered no, but over time it became part of Newtonianism that the right kind of mathematical law *does count* as an explanation. It is Kuhn’s view that the idea of explanation will evolve as our ideas about science and about the universe change.

So although the covering law theory definitely fails as a general account of explanation in science, it would be a mistake to conclude that *no* explanations have the form described by the covering law theory. There are some explanations that are at least close to what Hempel had in mind. The mistake is to apply that model to all cases.

I suggest that Kuhn was right on this point. I add that this proposal need not lead to the radical idea that *anything* can count as an explanation. Scientific traditions will generally have good reasons for their treatment of the idea of explanation; views about explanation will depend on views about what the world contains, for example. Some possible concepts of explanation will embed factual errors. To use a simple example, if someone claims that good explanations are always based on a concept of God’s will, but it turns out that there is no God, then that conception of explanation will be mistaken because of a factual error. Some philosophers might make the same argument about concepts of explanation that use the idea of causation—they might argue that the traditional idea of causal connection is a piece of mistaken metaphysics. But many possible treatments of the idea of explanation will not be ruled out by factual errors.

This is a case where it is important to pay attention to the actual use of the term “explain” in science. Here we find a lot of diversity. In some fields, there are technical senses of the word, even mathematical measurements of “the amount of variation explained” by a given factor. In other fields, nothing like a technical standard applies. The word “explain” also has an almost rhetorical use. Someone might say: “your theory does accommodate this result, but it does not really explain it.” This might mean “your theory can only be used to derive this result in an unnatural-looking way.” (Often, unification seems important in cases like this.) At other times the word “explain” is used in a much more low-key way in science. According

to scientific realism, a lot of science is aimed at describing what is going on in the world; often this will be a matter of describing *how things work*. How does photosynthesis work? How does the replication of DNA work? Descriptions of phenomena of this kind will often be referred to as explanations, but this does not mean that something *extra* is going on, beyond the description of mechanisms and processes.

At this point I should compare my view with another unorthodox position in this area, that of van Fraassen (1980). He denies, as I do, that explanation is some single, special relation common to all of science. He has developed a “pragmatic” account of explanation, in which what counts as an explanation varies according to context. But this is very different from my view. Van Fraassen wants to deny that explanation is something “inside” science at all; he denies that scientific reasoning includes the assessment of the explanatory power of theories. Instead, explanation is something that people do when they take scientific theories and use them to answer questions that are external to scientific discussion itself. In contrast, the view I am defending is a view in which explanation is thoroughly internal to science, *but variously so*. Assessments of what explains what are a very important part of scientific reasoning, but different fields may use somewhat different concepts and standards of explanation.

Before leaving this topic, I should also add a comment about *explanatory inference*. Back in chapter 3, I used this term for inference from a set of data to a hypothesis about a structure or process that would explain the data. In chapter 14, when I look at more recent discussions of confirmation and evidence, I will return to this topic. The term “explanatory inference” suggests that there is one kind of relationship between data and hypothesis—the explanation relationship—that is involved in explanatory inference. Many philosophers would accept this. The term “inference to the best explanation” is, in fact, a more common name for what I call explanatory inference; that term suggests that there is some single measure of “explanatory goodness” involved. But I think this is the wrong way to think about scientific reasoning (and this is why I have avoided the term “inference to the best explanation”). I use “explanatory inference” in a broader way that does not suppose that there is a single measure of explanatory goodness involved, which applies to all of science. Rather, explanatory inference is a matter of devising and comparing hypotheses about hidden structures that might be responsible for data. “Explanation” is seen as something pretty diverse.

To sum up: the covering law theory is dead, as a general account of explanation in science. But we should not look for a new theory of some single relation or pattern that is involved in all scientific explanation. Very often,

causation is involved. The same goes for unification and for deriving phenomena from laws. But different fields have different concepts and standards of explanation.

13.4 Laws and Causes (Optional Section)

This short and rather abstract section is a digression from the main themes of the book, a foray into an intersection between philosophy of science and the controversial field of *metaphysics*. The covering law theory of explanation made use of the idea of a law of nature. One of the theories that replaced it made use of the idea of causation. But what are laws of nature? What is causation?

In both cases, we have concepts that seem aimed at picking out a special kind of *connection* between things in the world. Causation is sometimes called, half jokingly, “the cement of the universe” (Mackie 1980; the phrase was first used by Hume [(1740) 1978]). In recent years, many philosophers have been skeptical about these concepts. But generally, their attitude has not been to reject them (“There is no such thing as causation”) but instead to reconstrue these concepts in a very low-key way (“Yes, there is causation, but it is no more than this . . .”). In particular, philosophers have tried to analyze both laws and causation in terms of patterns in the *arrangement* of things, rather than some extra connection *between* things. Sometimes this project is referred to as “Humeanism,” after David Hume, the first philosopher to develop a really focused suspicion about concepts of connection between events in nature (see also Lewis 1986b). The present-day Humeans do not have the same kind of empiricism as Hume, but they do want to avoid believing in any sort of unobservable cement connecting the universe together.

So a philosopher with Humean views will try to construe laws of nature as no more than regularities, or basic patterns, in the arrangement of events. To treat laws of nature in this way is to leave behind one of the familiar connotations of the term “law.” Usually, we see laws as directing, or guiding, or governing in some way. It is possible (indeed traditional) to see laws of nature as governing the flow of events in the universe. Laws are seen as *responsible* for the regular patterns that we see, rather than being *identical* with those patterns (Dretske 1977; Armstrong 1983). The Humean regards this “governing” conception of laws as a seduction that must be avoided by hard-headed philosophers. The logical empiricist attitude toward laws of nature was basically Humean, in this sense.

The topic of causation has generated a similar debate. On one side we have those who see causation as basically some special kind of regular pat-

tern in the flow of events. On the other side are those who see causation as a connection between things that is somehow *responsible* for the patterns (see Sosa and Tooley 1993). Perhaps this connection need not be seen as a mysterious philosophical entity; maybe it can be described by ordinary science (Dowe 1992; Menzies 1996).

For some years philosophers tended to discuss laws and their role in science in a way that had little contact with actual scientific work. In 1983 Nancy Cartwright delivered a wake-up call to the field with a book called *How the Laws of Physics Lie*, in which she argued that what people call “laws of physics” do not usually describe the behavior of real systems at all, but only describe the behavior of highly idealized fictional systems. Another important change that resulted from a closer look at actual science is that philosophers are no longer obsessed with natural laws as *the* goal of scientific theorizing. Over many years philosophers searched fields like biology for statements of laws of nature. Philosophers thought that any genuine science had to contain hypothesized laws and had to organize its ideas via the concept of a law. In fact, most biology has little use for the concept of a law of nature, but that does not make it any less scientific.

Further Reading

Wesley Salmon’s *Four Decades of Scientific Explanation* (1989) is a very good survey of work on explanation between 1948 (the advent of the covering law theory) and the late 1980s. (The only thing marring Salmon’s discussion is his rather eccentric theory about causation, which affects his treatment of explanation.)

A very good alternative discussion of causation and explanation can be found in Lewis 1986a. (Lewis’s theory of causation is also eccentric. In fact, I guess *every* philosopher’s theory of causation is eccentric; no two philosophers seem to agree. Lewis’s discussion is compatible with a range of different views about causation, though.)

For unificationist theories of explanation, see Friedman 1974 and Kitcher 1989. These are fairly advanced papers.

Lewis discusses the Humean program in metaphysics in the preface to his 1986 *Philosophical Papers*, volume 2. Armstrong 1983 is a clear introduction to the more purely philosophical side of the literature on laws of nature. Beebe 2000 is a good discussion of the idea that laws “govern” things. Mitchell (2000) defends an interesting position on laws.